

## 13.4 Bonferroni and Holm

Scheffé's and Tukey's methods need an exact distribution for the worst error in a family. The methods of this section need only the level of each separate test and a probability inequality, so they apply to any finite list of hypotheses in any model, whatever the dependence. Holm's step-down procedure recovers part of the resulting conservatism at no cost in generality.

### 13.4.1 The Bonferroni inequality

#### Theorem 13.4.1 — Bonferroni

- (a) For any events  $A_1, \dots, A_m$ ,  $\Pr(\bigcup_j A_j) \leq \sum_j \Pr(A_j)$ .
- (b) Let  $\alpha_1, \dots, \alpha_m > 0$  with  $\sum_j \alpha_j = \alpha$ . If  $[L_j, U_j]$  is a confidence interval for  $\psi_j$  with coverage at least  $1 - \alpha_j$ , for each  $j$ , then the  $m$  intervals have simultaneous coverage at least  $1 - \alpha$ .
- (c) Let  $p_j$  be a  $p$ -value for  $H_j$ , valid in the sense that  $\Pr(p_j \leq u) \leq u$  for  $0 \leq u \leq 1$  whenever  $H_j$  is true. The procedure that rejects  $H_j$  when  $p_j \leq \alpha_j$  has PFER  $\leq \sum_{j \in I_0} \alpha_j \leq \alpha$ , and hence controls the familywise error rate strongly at level  $\alpha$ .

No assumption is made about the dependence among the events, intervals or  $p$ -values.

*Proof.* (a) Pointwise, the indicator of a union is at most the sum of the indicators:  $1_{\bigcup_j A_j} \leq \sum_j 1_{A_j}$ , since the right side is at least 1 wherever the left side is. Take expectations. (b) Apply (a) to the events  $A_j = \{\psi_j \notin [L_j, U_j]\}$ . (c) The number of false rejections is  $V = \sum_{j \in I_0} 1_{\{p_j \leq \alpha_j\}}$ , so  $E(V) = \sum_{j \in I_0} \Pr(p_j \leq \alpha_j) \leq \sum_{j \in I_0} \alpha_j$ . Strong control follows from Proposition 13.1.1. ■

Part (c) bounds even the expected *number* of false rejections, which is why the procedure is conservative when false rejections tend to occur together.

In the linear model, the **Bonferroni intervals** for  $k$  estimable functions  $\lambda_1^\top \beta, \dots, \lambda_k^\top \beta$  chosen in advance are

$$\lambda_j^\top \hat{\beta} \pm t_{\nu, \alpha/(2k)} \text{se}(\lambda_j^\top \hat{\beta}), \quad j = 1, \dots, k, \quad (13.4.1)$$

each the  $t$  interval of Theorem 12.1.1 at level  $1 - \alpha/k$ . The equal split is conventional; any split fixed before the data are seen is valid.

*Warning.* The Bonferroni guarantee is for a list of  $k$  hypotheses fixed in advance. If the analyst looks at the estimates and then chooses which  $k$  to test, the hypotheses actually tested are a data-dependent selection from a larger family, and the relevant  $k$  is the size of that larger family. For data-driven choices within a subspace, use Scheffé's method.

## 13.4.2 Independent tests: Šidák's improvement

If the tests of the true hypotheses are independent and each has exact level  $\alpha$ , then by Eq. 13.1.1 the familywise error rate is  $1 - (1 - \alpha)^{m_0}$ . Choosing

$$\alpha = 1 - (1 - \alpha)^{1/m} \quad (13.4.2)$$

makes this at most  $1 - (1 - \alpha)^m = \alpha$ , with equality when all hypotheses are true. This **Šidák level** is slightly larger than  $\alpha/m$ , because  $(1 - \alpha/m)^m > 1 - \alpha$  for  $m \geq 2$  (Bernoulli's inequality), so the Šidák procedure is slightly less conservative than Bonferroni's.

In a linear model, however,  $t$  statistics are never independent: they share  $s$ . The following result shows that the sharing works in the right direction.

### Lemma 13.4.2 — Chebyshev's association inequality

Let  $U$  be a real random variable and  $f_1, \dots, f_k$  nonnegative, nondecreasing functions with  $E f_j(U)^k < \infty$ . Then  $E[\prod_j f_j(U)] \geq \prod_j E f_j(U)$ .

*Proof.* For two functions  $f, h$ , let  $U'$  be an independent copy of  $U$ . Since  $f$  and  $h$  are both nondecreasing,  $(f(U) - f(U'))(h(U) - h(U')) \geq 0$ . Taking expectations gives  $2E[f(U)h(U)] - 2E f(U) E h(U) \geq 0$ . For  $k$  functions, argue by induction: the product  $f_1 \cdots f_{k-1}$  of nonnegative nondecreasing functions is nonnegative and nondecreasing, so  $E[f_1 \cdots f_k(U)] \geq E[f_1 \cdots f_{k-1}(U)] E f_k(U)$ , and the induction hypothesis applies to the first factor. The moment condition makes every expectation finite, by Hölder's inequality. ■

### Proposition 13.4.3 — Šidák intervals

Let  $Z_1, \dots, Z_k$  be independent  $N(0, 1)$  variables and  $V \sim \chi^2(\nu)$  independent of them, and put  $T_j = Z_j / \sqrt{V/\nu}$ . Then for every  $c > 0$ ,

$$\Pr(|T_j| \leq c \text{ for all } j) \geq \prod_{j=1}^k \Pr(|T_j| \leq c).$$

Consequently, if  $\lambda_1^\top \hat{\beta}, \dots, \lambda_k^\top \hat{\beta}$  are uncorrelated, the intervals  $\lambda_j^\top \hat{\beta} \pm c \text{se}(\lambda_j^\top \hat{\beta})$  with  $c$  the upper  $\frac{1}{2}(1 - (1 - \alpha)^{1/k})$  point of  $t(\nu)$  have simultaneous coverage at least  $1 - \alpha$ .

*Proof.* Conditionally on  $V = v$ , the events  $\{|Z_j| \leq c\sqrt{v/\nu}\}$  are independent, each with probability  $f(v) = 2\Phi(c\sqrt{v/\nu}) - 1$ . So  $\Pr(|T_j| \leq c \forall j) = E[f(V)^k]$ , while  $\Pr(|T_j| \leq c) = E f(V)$ . The function  $f$  is nonnegative, bounded and nondecreasing, and Lemma 13.4.2 gives  $E[f(V)^k] \geq (E f(V))^k$ . For the intervals, the estimates are jointly normal (Theorem 7.5.1), so uncorrelated estimates are independent (Theorem 3.4.1), and they are independent of  $s$ . The standardized errors are therefore of the form  $T_j$ ,

and each  $\Pr(|T_j| \leq c) = (1 - \alpha)^{1/k}$ . ■

The proposition covers orthogonal contrasts (Proposition 8.7.2(c)) and regressions with orthogonal columns. Šidák (1967) proved the corresponding inequality  $\Pr(|Z_j| \leq c_j \forall j) \geq \prod_j \Pr(|Z_j| \leq c_j)$  for *any* centred jointly normal  $(Z_1, \dots, Z_k)$ ; with the conditioning argument above it makes Šidák's intervals valid for any finite family of estimable functions. We use that theorem without proof. The gain over Bonferroni is small: for  $k = 10$  and  $\nu = 29$  the multipliers are 3.0289 and 3.0380, and with independent numerators and a shared  $s$  the exact coverage of the nominal 95% intervals, computed by integrating over  $s$ , is 0.9520 and 0.9530.

### 13.4.3 The closure principle

Bonferroni's procedure keeps the threshold  $\alpha/m$  even after rejections, although a rejected false hypothesis can no longer be falsely rejected. The closure principle exploits this. For each nonempty  $I \subseteq \{1, \dots, m\}$  let  $H_I = \bigcap_{i \in I} H_i$  be the **intersection hypothesis**, and let  $\phi_I$  be a level- $\alpha$  **local test** of it.

#### Theorem 13.4.4 — Closed testing

The procedure that rejects  $H_j$  iff  $\phi_I$  rejects  $H_I$  for every  $I$  containing  $j$  controls the familywise error rate strongly at level  $\alpha$ .

*Proof.* Fix the parameters and suppose  $I_0 \neq \emptyset$ ; otherwise there is nothing to prove. The intersection hypothesis  $H_{I_0}$  is true, so  $\Pr(\phi_{I_0} \text{ rejects}) \leq \alpha$ . If some true  $H_j$  is rejected, then  $j \in I_0$ , so by the definition of the procedure  $\phi_{I_0}$  rejects. Hence  $\Pr(V \geq 1) \leq \Pr(\phi_{I_0} \text{ rejects}) \leq \alpha$ . ■

Rejecting one hypothesis requires rejecting every intersection containing it, in general  $2^{m-1}$  tests. Holm's procedure is a case where this collapses to  $m$  steps.

### 13.4.4 Holm's step-down procedure

Order the  $p$ -values as  $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$ , breaking ties arbitrarily, and let  $H_{(i)}$  be the hypothesis with the  $i$ th smallest  $p$ -value.

**Holm's procedure at level  $\alpha$ .** For  $i = 1, 2, \dots, m$  in turn: if  $p_{(i)} \leq \alpha/(m - i + 1)$ , reject  $H_{(i)}$  and continue; otherwise stop, and reject none of  $H_{(i)}, \dots, H_{(m)}$ .

The first threshold is Bonferroni's,  $\alpha/m$ . Each rejection relaxes the next threshold: to  $\alpha/(m-1)$ , then  $\alpha/(m-2)$ , and so on, up to  $\alpha$  for the last hypothesis.

#### Theorem 13.4.5 — Holm

Let each  $p_j$  be valid for  $H_j$ , as in [Theorem 13.4.1\(c\)](#), with any dependence among them. Then:

- (a) Holm's procedure controls the familywise error rate strongly at level  $\alpha$ ;
- (b) it rejects every hypothesis that Bonferroni's procedure (all  $\alpha_j = \alpha/m$ ) rejects, and possibly more.

*Proof.* (a) Suppose  $m_0 \geq 1$ , and let  $j^*$  be the rank, in the ordering of all  $m$   $p$ -values, of the smallest  $p$ -value belonging to a true hypothesis, so that  $p_{(j^*)} = \min_{i \in I_0} p_i$ . The hypotheses ranked before  $j^*$  are all false, and there are only  $m - m_0$  false hypotheses, so  $j^* \leq m - m_0 + 1$  and  $\alpha/(m - j^* + 1) \leq \alpha/m_0$ . Holm's procedure rejects an initial segment  $H_{(1)}, \dots, H_{(i)}$  of the ordering. If that segment contains any true hypothesis, it contains  $H_{(j^*)}$ , so  $p_{(j^*)} \leq \alpha/(m - j^* + 1) \leq \alpha/m_0$ . Therefore

$$\Pr(V \geq 1) \leq \Pr\left(\min_{i \in I_0} p_i \leq \frac{\alpha}{m_0}\right) \leq \sum_{i \in I_0} \Pr\left(p_i \leq \frac{\alpha}{m_0}\right) \leq \alpha,$$

by [Theorem 13.4.1\(a\)](#) and the validity of the  $p$ -values. (b) If Bonferroni rejects  $H_{(i)}$ , then  $p_{(j)} \leq p_{(i)} \leq \alpha/m \leq \alpha/(m - j + 1)$  for every  $j \leq i$ , so Holm's procedure does not stop before step  $i$  and rejects  $H_{(i)}$ . ■

Bonferroni's  $\alpha/m$  pays for  $m$  potential errors when only  $m_0$  are possible. Holm's procedure is closed testing with the Bonferroni local tests ([Exercise \(B3\)](#)). It gives no simple simultaneous intervals, since the threshold for a hypothesis depends on the other  $p$ -values; its output is a list of rejections, or **adjusted  $p$ -values**

$$\tilde{p}_{(i)} = \max_{j \leq i} \min\{1, (m - j + 1) p_{(j)}\}, \quad (13.4.3)$$

with  $H_{(i)}$  rejected at level  $\alpha$  iff  $\tilde{p}_{(i)} \leq \alpha$  ([Exercise \(B1\)](#)).

#### Example. (Holm for the education pairs)

For the 21 pairwise comparisons of Example (All pairs of education levels) the smallest  $p$ -value is  $1.1 \times 10^{-5}$ . Bonferroni's threshold is  $0.05/21 = 0.00238$ , and 8 of the  $p$ -values fall below it. Holm's procedure starts at the same threshold but relaxes it after each rejection. It rejects 10 hypotheses: the tenth smallest  $p$ -value is 0.0037, below its threshold  $0.05/12$ , and the eleventh, 0.0126, exceeds  $0.05/11 = 0.0045$ , so the procedure stops.

```
import itertools
import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
```

```

means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                     # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu      # pooled variance estimate
s = np.sqrt(s2)
alpha = 0.05
pairs = list(itertools.combinations(range(g), 2))
diff = np.array([means[k] - means[l] for k, l in pairs])
se_d = np.array([s * np.sqrt(1 / n_k[k] + 1 / n_k[l]) for k, l in pairs])
t = diff / se_d
pval = 2 * stats.t.sf(np.abs(t), nu)

def holm(p, alpha):
    """Holm's step-down procedure: reject the i smallest p-values, i maximal with
    p_(j) <= alpha / (m - j + 1) for all j <= i."""
    m = len(p)
    order = np.argsort(p)
    reject = np.zeros(m, dtype=bool)
    for j, idx in enumerate(order):           # j = 0, 1, ...
        if p[idx] > alpha / (m - j):
            break
        reject[idx] = True
    return reject

def benjamini_hochberg(p, q):
    """Reject the k smallest p-values, k the largest index with p_(k) <= k q / m."""
    m = len(p)
    order = np.argsort(p)
    below = np.nonzero(p[order] <= q * np.arange(1, m + 1) / m)[0]
    reject = np.zeros(m, dtype=bool)
    if below.size:
        reject[order[: below[-1] + 1]] = True
    return reject

print("Holm rejects", holm(pval, alpha).sum(), " BH (q = 0.05) rejects",
      benjamini_hochberg(pval, alpha).sum())

```

[Run this code online](#)

### 13.4.5 How much does Holm gain?

If all hypotheses are true, Holm and Bonferroni make a false rejection in the same data sets. The more hypotheses are false and rejected early, the larger the later thresholds and the gain.

#### Example. (Ten regression coefficients)

A regression has an intercept and ten centred, mutually orthogonal regressors, with  $n = 40$ . The ten  $t$  statistics for the slopes are then  $T_j = Z_j / \sqrt{V/29}$  with independent  $Z_j \sim N(\delta_j, 1)$  and a shared  $V \sim \chi^2(29)$  (Theorem 7.5.1). Let  $k$  of the slopes be nonzero, each with  $\delta_j = 3.5$ , and test all ten at familywise level 0.05. [Figure 13.4.1](#) shows the result of 20000 simulated data sets for each  $k$ . All three procedures keep the familywise error rate at or below 0.05, up to simulation error. The average power of Bonferroni's procedure, the proportion of nonzero slopes it detects, hardly depends on  $k$ : it is 0.675 with one nonzero slope and 0.676 with five or nine. Šidák's is larger by

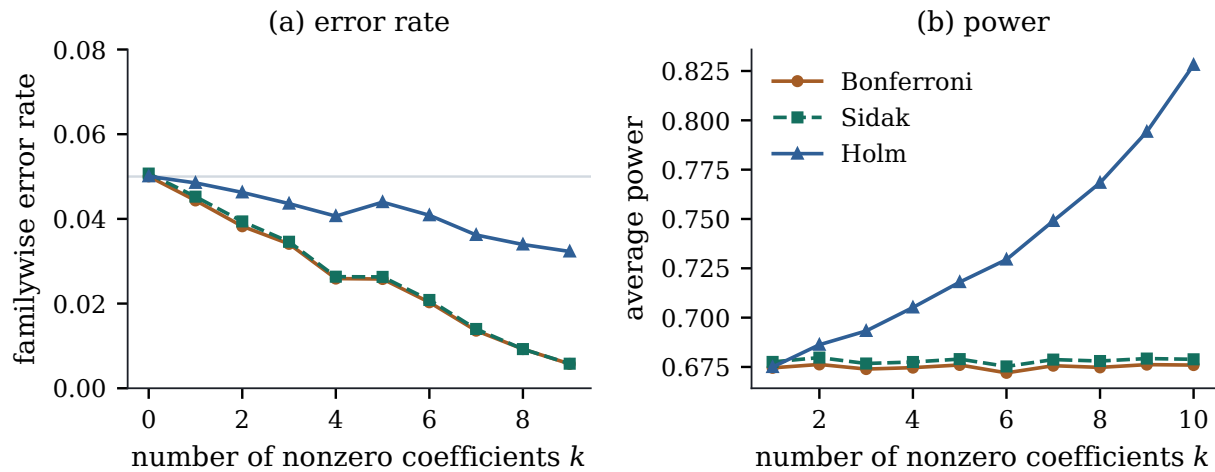


Figure 13.4.1: Ten orthogonal regression coefficients,  $k$  of them nonzero, tested at familywise level 0.05: (a) familywise error rate, (b) average power, the share of nonzero coefficients detected.

a few thousandths. Holm's power is 0.675 with one nonzero slope, 0.718 with five and 0.794 with nine.

```
import numpy as np
from scipy import stats

alpha, m, nu, delta, reps = 0.05, 10, 29, 3.5, 20_000

def simulate(k, reps, seed):
    """FWER and average power of three procedures when k of m coefficients are
    nonzero."""
    rng = np.random.default_rng(seed)
    shift = np.r_[np.full(k, delta), np.zeros(m - k)]
    T = (rng.normal(size=(reps, m)) + shift) / np.sqrt(rng.chisquare(nu, (reps, 1))
    / nu)
    p = 2 * stats.t.sf(np.abs(T), nu)
    rules = {
        "Bonferroni": p <= alpha / m,
        "Sidak": p <= 1 - (1 - alpha) ** (1 / m),
    }
    # Holm: sort each row, compare p_(j) with alpha / (m - j + 1), stop at the first
    failure
    order = np.argsort(p, axis=1)
    ps = np.take_along_axis(p, order, axis=1)
    passed = np.cumprod(ps <= alpha / (m - np.arange(m)), axis=1).astype(bool)
    holm = np.zeros_like(passed)
    np.put_along_axis(holm, order, passed, axis=1)
    rules["Holm"] = holm
    out = {}
    for name, rej in rules.items():
        fwer = np.mean(rej[:, k:].any(axis=1)) if k < m else np.nan
        power = rej[:, :k].mean() if k > 0 else np.nan
        out[name] = (fwer, power, rej)
    return out

res = simulate(5, reps, seed=5)
for name, (fwer, power, _) in res.items():
    print(f"{name:10s} FWER = {fwer:.3f}    average power = {power:.3f}")
```

[Run this code online](#)

There is no reason to use Bonferroni's *tests* rather than Holm's, though Bonferroni's *intervals* remain useful. With independent  $p$ -values, Hochberg (1988) proposed the step-up version: reject  $H_{(1)}, \dots, H_{(i)}$  for the largest  $i$  with  $p_{(i)} \leq \alpha/(m - i + 1)$ . It rejects everything Holm's procedure does ([Exercise \(C1\)](#)), but its validity rests on an inequality of Simes (1986) that holds under independence and some positive dependence, not in general. The step-up idea returns in the next section, controlling a different error rate.

## 13.4.6 Exercises

### 13.4.6.1 A. Check your understanding

#### Exercise A1.

Five hypotheses have  $p$ -values 0.001, 0.011, 0.012, 0.04, 0.2. At familywise level 0.05, which are rejected by Bonferroni's procedure and which by Holm's? Compute Holm's adjusted  $p$ -values [Eq. 13.4.3](#).

*Solution.* Bonferroni's threshold is 0.01, so only the first is rejected. Holm compares  $0.001 \leq 0.05/5$ ,  $0.011 \leq 0.05/4$ ,  $0.012 \leq 0.05/3$ , and then  $0.04 > 0.05/2$ , so it rejects the first three and stops. The adjusted  $p$ -values are  $5 \times 0.001 = 0.005$ ,  $\max(0.005, 4 \times 0.011) = 0.044$ ,  $\max(0.044, 3 \times 0.012) = 0.044$ ,  $\max(0.044, 2 \times 0.04) = 0.08$  and  $\max(0.08, 0.2) = 0.2$ . The three below 0.05 are the rejected hypotheses.

### 13.4.6.2 B. Practice

#### Exercise B1.

Show that Holm's procedure rejects  $H_{(i)}$  iff the adjusted  $p$ -value [Eq. 13.4.3](#) satisfies  $\tilde{p}_{(i)} \leq \alpha$ , and that the adjusted  $p$ -values are nondecreasing in  $i$ .

#### Exercise B2.

All  $m \geq 2$  hypotheses are true and the  $p$ -values are independent and uniform. Show that the per-family error rate of Bonferroni's procedure is exactly  $\alpha$ , while that of Holm's procedure is larger than  $\alpha$ . So Holm's procedure controls the familywise rate but not the per-family rate.

#### Exercise B3.

Show that Holm's procedure is closed testing ([Theorem 13.4.4](#)) with the Bonferroni local tests  $\phi_I$ : reject  $H_I$  when  $\min_{i \in I} p_i \leq \alpha/|I|$ .

*Solution.* Suppose Holm rejects  $H_{(1)}, \dots, H_{(i)}$  and stops at step  $i + 1$ . Take  $j \leq i$  and any  $I \ni (j)$ , writing  $(j)$  for the index of  $H_{(j)}$ . If  $I$  contains some  $(l)$  with  $l \leq j$ , let  $l$  be the smallest such; then  $I$  contains no  $(1), \dots, (l - 1)$ , so  $|I| \leq m - l + 1$  and  $\min_I p \leq p_{(l)} \leq \alpha/(m - l + 1) \leq \alpha/|I|$ :  $\phi_I$  rejects. So closed testing rejects  $H_{(j)}$ . Conversely, the

set  $I = \{(i+1), \dots, (m)\}$  has  $m-i$  elements and minimum  $p_{(i+1)} > \alpha/(m-i)$ , so  $\phi_I$  does not reject, and no hypothesis in  $I$  is rejected by closed testing. The two procedures therefore reject the same hypotheses.

### 13.4.6.3 C. Going deeper

#### Exercise C1.

Show that Hochberg's step-up procedure rejects every hypothesis rejected by Holm's step-down procedure with the same thresholds. Give an example with  $m = 2$  where Hochberg rejects both hypotheses and Holm rejects neither.